

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – EXPERIMENTATION

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – EXPERIMENTATION
Weightage:	30 % of CIA	Total Marks:	20 marks
CO2 Statement:	<i>Compare mobile communication technologies, including multiple access techniques and network evolution, based on performance and applications.</i>		
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Section / Batch:	Slot D	Date:	31-07-2026

Code of Conduct

I S Sandhiya (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains ONE question.
- Duration: 30 minutes.

Annexure A - QUESTIONS

1. Figma-Based Mobile IP Communication Workflow Analysis (BL4 – Analyze)

Design an interactive Figma prototype illustrating the complete workflow of Mobile IP communication when a Mobile Node moves from its Home Network to a Foreign Network. Your design should include the Home Agent (HA), Foreign Agent (FA), Correspondent Node (CN), Care-of Address (CoA), Registration Process, and Packet Tunneling.

Analyze the communication process by identifying:

- The purpose of each network component.
- Points where packet delay or packet loss may occur.
- The effect of node mobility on communication performance.
- Possible improvements to reduce latency.

Deliverables:

- Figma (.fig) design
- GitHub repository containing exported images/PDF and documentation (README).

Solution:

Figma-Based Mobile IP Communication Workflow Analysis

Aim

To design and analyze the complete Mobile IP communication workflow using an interactive Figma prototype. The workflow illustrates how a Mobile Node (MN) moves from its Home Network to a Foreign Network while maintaining uninterrupted communication through Mobile IP.

Introduction

Mobile IP is a communication protocol that allows a Mobile Node (MN) to move from one network to another without changing its permanent IP address. It enables continuous communication by using a Home Agent (HA), Foreign Agent (FA), Care-of Address (CoA), and packet tunneling mechanism. This ensures that ongoing connections remain active even when the mobile device changes its location.

Mobile IP Communication Workflow

Step 1: Mobile Node in Home Network

Initially, the Mobile Node (MN) is connected to its Home Network and communicates normally with the Correspondent Node (CN). The Home Agent (HA) keeps track of the Mobile Node's permanent IP address.

Step 2: Mobile Node Moves to Foreign Network

When the Mobile Node moves to a Foreign Network, it connects with the Foreign Agent (FA). The Foreign Agent provides a temporary Care-of Address (CoA), which represents the Mobile Node's current location.

Step 3: Registration Process

The Mobile Node sends a Registration Request containing its Care-of Address to the Home Agent through the Foreign Agent. The Home Agent updates its mobility binding table and sends a Registration Reply confirming the registration.

Step 4: Packet Tunneling

The Correspondent Node continues sending data packets to the Mobile Node's permanent Home Address. The Home Agent intercepts these packets and tunnels them to the Care-of Address. The Foreign Agent receives the tunneled packets and forwards them to the Mobile Node.

Step 5: Communication Continues

The Mobile Node receives the packets successfully while staying connected to the Foreign Network. The communication continues without changing the Mobile Node's permanent IP address.

Purpose of Each Network Component

Mobile Node (MN)

The Mobile Node is the device that moves between different networks while maintaining continuous communication using its permanent IP address.

Home Agent (HA)

The Home Agent maintains the Mobile Node's permanent IP address and forwards packets to its current Care-of Address through packet tunneling.

Foreign Agent (FA)

The Foreign Agent provides the Care-of Address to the Mobile Node, assists in the registration process, and delivers tunneled packets to the Mobile Node.

Correspondent Node (CN)

The Correspondent Node is the communicating device that exchanges data with the Mobile Node. It is unaware of the Mobile Node's current location.

Care-of Address (CoA)

The Care-of Address is a temporary IP address assigned to the Mobile Node when it connects to a Foreign Network. It represents the Mobile Node's current location.

Analysis

1. Purpose of Each Network Component

Component	Purpose
Mobile Node (MN)	Moves between networks while maintaining communication
Home Agent (HA)	Stores permanent address and tunnels packets
Foreign Agent (FA)	Assigns Care-of Address and forwards packets
Correspondent Node (CN)	Sends and receives data packets
Care-of Address (CoA)	Represents the Mobile Node's current location

2. Points Where Packet Delay or Packet Loss May Occur

Packet Delay

Packet delay may occur during:

- Registration between the Mobile Node and Home Agent.
- Packet tunneling from Home Agent to Foreign Agent.
- Wireless transmission in the Foreign Network.
- Network congestion during data transfer.

Packet Loss

Packet loss may occur due to:

- Handover between Home Network and Foreign Network.
- Weak wireless signal.
- Network congestion.
- Registration failure.
- Tunnel interruption.

3. Effect of Node Mobility on Communication Performance

When the Mobile Node frequently changes its location:

- Communication latency increases.
- Additional signaling overhead is generated.
- Temporary packet loss may occur.
- Delay increases during handoff.
- Overall communication performance may decrease if mobility is frequent.

4. Possible Improvements to Reduce Latency

The following techniques can improve Mobile IP performance:

- Fast Mobile IP to reduce handoff delay.
- Route Optimization to avoid unnecessary packet tunneling.
- Hierarchical Mobile IP for faster registration.
- Efficient tunneling mechanisms.
- Better wireless network coverage.
- Faster registration process.
- Improved mobility management algorithms.

Advantages

- Supports seamless mobility.
- Maintains ongoing communication.
- Permanent IP address remains unchanged.
- Enables uninterrupted internet connectivity.
- Supports roaming users efficiently.

Disadvantages

- Packet tunneling increases delay.
- Additional signaling overhead.
- Registration process consumes time.
- Higher packet loss during frequent movement.

- Increased complexity in network management.

Conclusion

The Mobile IP communication workflow enables uninterrupted communication by allowing a Mobile Node to move between different networks while keeping its permanent IP address unchanged. The Home Agent, Foreign Agent, Care-of Address, registration process, and packet tunneling work together to ensure seamless connectivity. Although mobility introduces packet delay and signaling overhead, techniques such as Fast Mobile IP, Route Optimization, and Hierarchical Mobile IP can significantly reduce latency and improve communication performance.